Abelian Covers and Gale Correspondences: Rational Support Filters, Smaller Algebraic Sources, and Exact Determinant Certificates

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Abstract

We give a scoped mathematical assessment of the September 30 source on abelian covers, Gale correspondences, and Hodge classes on powers. Its global statements concern tensor Hodge-generic points of a full marked variation, not every period or every restricted family. We separate those statements from finite algebraic constructions. Rational support and exact-order projectors give explicit complete-orbit blocks, rank formulas, and determinant extraction on curve powers. In the Gale supply, a single point replaces an entire finite dual torsor; full-support cohomology uses only a specified isogeny factor of the dual Jacobian. Character-order reduction further replaces individual blocks by smaller-degree Fermat varieties and cyclic quotient curves. A bounded-source assembly yields a precise weaker sufficient Hodge hypothesis on powers of one fixed product. The signed complementary-wedge map supplies a normalization-sensitive certificate, not a globally unitary period frame. These are local correspondence and source-economy refinements, with independently executable exact checks. They neither verify the imported all-genus monodromy and CM Hodge theorems nor establish a new general Hodge conjecture theorem. Contribution assessment: C3/5, notable source-specific reductions of algebraic supply and exact certification, with the global geometric and Hodge inputs explicitly retained.

1 Scope, conventions, and the source architecture

All varieties below are over $\mathbb{C}$. Cohomology is singular cohomology with rational coefficients unless a cyclotomic extension E is displayed. "Rational correspondence" means a cycle with rational coefficients on the fixed complex varieties; it does not mean a cycle defined over $\mathbb{Q}$. Our convention is $V(j)^{p,q} = V^{p+j,q+j}$, of weight $\mathrm{wt}(V) - 2j$. For smooth projective T, X, a cycle $\Gamma \in \mathrm{CH}^{\dim T+j}(T \times X)_{\mathbb{Q}}$ induces

$$\Gamma_* : H^{k-2j}(T, \mathbb{Q})(-j) \longrightarrow H^k(X, \mathbb{Q}). \tag{1.1}$$

Automorphism graphs are chosen to act by $\rho(g) = (g^{-1})^*$, a left action. In the coordinate-root part we instead use the source's commuting pullbacks g^*; replacing a character by its inverse changes none of the support or orbit assertions. We specify the sign when contraction matters.

The source [1] fixes a connected topological abelian cover $\phi : \pi_1(B \setminus D, *) \twoheadrightarrow P$, with active branch meridians, allowing any base genus, no branching, and the trivial group. It varies all marked

base complex structures and branch positions. Its claim is, in order,

$$\forall \phi \quad \forall s \in \mathcal{T}_{\phi}^{\mathrm{Hg}} \quad \forall M \text{ CM} \quad \forall N \geq 1 : \mathrm{HC}\big((\mathrm{Jac}(C_s) \times M)^N\big).$$

The same locus is chosen before the CM companion. Here tensor genericity means that any rational balanced tensor Hodge at s is Hodge throughout the full marked variation, in every tensor degree. This is not a test of finitely many periods. For the diagonal application the claim is very general in the full smooth labelled coefficient locus, for at most two equations of one common degree. The source removes no exceptional subset for one equation or dimension zero. Its quadratic parity remark already recognizes the absence of a full-support character when the number of coordinates is odd; we do not claim this observation as new.

Source component	Literal lines	Role and audit boundary
Statements and shared tools	1–467	Quantifiers; CM Hodge theorem and saturated tensor/transfer results imported from companions.
Determinant supply	468–1188	Quotient signs, Abel fiber, Fermat bundles, weight detection, inverse-character return.
Full marked variation	1189–2511	Rank-one twists, connected irreducibility, sphere links and saturation; read, not independently certified in full.
Root covers and Gale supply	2512–3063	Support ranks, actual graph, complementary wedges, degree shifts; principal local interface here.
Family comparison	3064–3515	Coordinate roots, based lifts, all marked fibers and algebraic Hodge loci; retained geometric obligations.
Assembly and references	3516–3779	Entire source degrees, products, low-dimensional cases and dependencies.

All 3779 lines of the public TeX and all 820 lines of the relevant private original [2] were literally read in bounded displays. This is not a whole-corpus read or a full proof audit. The private proof passages on exact cycle membership (241–287, 546–552), rational closure (554–585), lattice thresholds (587–616), and complete Galois-orbit projectors (627–656) are particularly relevant. The closure, Fourier, and perturbation principles there are inherited methods, not new results of this report. No private text or private title was sent to public search.

2 Actual algebraic projectors and determinant blocks

2.1 Finite quotients, including fixed points

For a finite algebraic action on a projective V, the quotient $q : V \to V/G$ satisfies $q^* H^i(V/G, \mathbb{Q}) = H^i(V, \mathbb{Q})^G$. Indeed proper base change makes $R^j q_* \mathbb{Q} = 0$ for $j > 0$, and each fiber is one orbit, so $(q_* \mathbb{Q})^G = \mathbb{Q}$. Rational averaging is exact and Leray gives the assertion. Pullback is a mixed-Hodge map; when V is smooth projective its invariant subspace is pure. Freeness is not required. A singular quotient does not thereby acquire an ordinary smooth Chow motive: for algebraic operations on V use the genuine graph projector

$$A_G = |G|^{-1} \sum_{g \in G} \rho(g) \in \mathrm{CH}^{\dim V}(V \times V)_{\mathbb{Q}}.$$

For a finite group quotient between smooth projective varieties, with the action generically free on every component and generic degree $|G|$, $q_* q^* = \deg(q)\,\mathrm{id}$ and $q^* q_* = \sum_g \rho(g)$ on rational cohomology. The degree is the generic degree, not the number of distinct points in a ramified

fiber. These equalities follow either from graph composition over the generic free locus with its multiplicities or from the transfer and projection formulas. We never substitute the order of an ineffective group for this degree.

Proposition 2.1 (Kernel-indexed rational projectors)**.** *Let a finite abelian P act algebraically on a smooth projective V. Fix a character χ of order $e > 1$, with kernel K, and identify $P/K = \mathbb{Z}/e$. For $g \in P$ let $a(g)$ be its image. Its full rational orbit projector is*

$$E_K = \frac{1}{|P|} \sum_{g \in P} c_e(a(g))\rho(g), \qquad c_e(a) = \sum_{d \mid \gcd(e,a)} d\,\mu(e/d). \tag{2.1}$$

It equals the sum of e_{χ^u} over $u \in (\mathbb{Z}/e)^\times$. Writing $A_L = |L|^{-1} \sum_{g \in L} \rho(g)$, an alternative with only divisor-indexed subgroup averages is

$$E_K = \sum_{d \mid e} \mu(e/d) A_{K_d}, \qquad K_d/K = d\mathbb{Z}/e. \tag{2.2}$$

Both are rational algebraic idempotents, not numerical projections. Characters with the same kernel form exactly this full Galois orbit.

Proof. The character idempotent is $|P|^{-1} \sum_g \chi(g)^{-1}\rho(g)$. Summing primitive powers gives the Ramanujan sum (unchanged on replacing a by $-a$). To derive its integer formula, insert $1_{(u,e)=1} = \sum_{k \mid (u,e)} \mu(k)$ into the exponential sum; the inner geometric sum is e/k when $e/k \mid a$ and zero otherwise. The average A_{K_d} selects characters of P/K of order dividing d; Möbius inversion gives exactly the order-e characters. Orthogonality proves idempotence. Every faithful character of a cyclic group of order e is a primitive power of any one; the automorphisms of $\mathbb{Q}(\zeta_e)$ realize all those powers. This also proves the kernel assertion. □

This is classical finite Fourier theory specialized to the source's cyclic quotients. The useful reduction is operational: one rational kernel block can replace $\varphi(e)$ separately conjugated objects, but its dimension is still the sum of their dimensions. Integer coefficients $c_e(a)$ do not eliminate the denominator $|P|$.

Proposition 2.2 (Determinants, not exterior powers of an orbit sum)**.** *Let $C \to B$ be a connected abelian cover and $\chi \neq 1$ have kernel K. Put $r = \dim H^1(C,E)_\chi = 2g(B) + r_\chi - 2 > 0$ and $G = P^r \rtimes \mathfrak{S}_r$. The product map $G \to P$ is a homomorphism. Let N_K be its inverse image of K. Averaging by N_K and then applying the faithful projector of $G/N_K \simeq P/K$ gives a rational algebraic projector $\mathcal{D}_K$ on C^r with*

$$\operatorname{im}(\mathcal{D}_K : H^r(C^r,E) \to H^r(C^r,E)) = \bigoplus_{u \in (\mathbb{Z}/e)^\times} \bigwedge^r H^1(C,E)_{\chi^u}. \tag{2.3}$$

This block has rational dimension $\varphi(e)$. No other cohomological degree survives. The rank-zero determinant is instead the unit on a point.

Proof. Product-kernel invariance forces the same character in every tensor slot; faithful projection forces that character to be χ^u. The connected curve has no nontrivial characters in H^0 or H^2, so only the r degree-one slots survive. Geometric permutations act there as their ordinary tensor permutations multiplied by their signs. Their invariants are precisely alternating tensors. The active inertia set is unchanged under primitive powers, so each character space has the same rank r. Its top exterior power is a line. All the operations are averages of automorphism graphs. The construction therefore proves the statement without asserting algebraicity of any vector in this block. □

In particular $\bigwedge^{r\varphi(e)}(E_K H^1)$ is not the block (2.3): it is a different, one-dimensional norm determinant in a different tensor degree. Nor is an unbalanced character determinant a Hodge class. The source supplies its full cohomology direction from CM sources and only later lifts rational Hodge classes; these distinctions are essential. Its inverse-character projector in the return pairing (source 1075–1098) is correct, as is its use of a resolving variety only for the correspondence, not as a claimed CM-supplied variety.

3 Support and exact-order filters for root covers

Fix $n \geq 2$, $m \geq 2$, and a coordinate-general $(d+1)$-plane $P \subset \mathbb{C}^m$, $0 \leq d \leq m-2$. Write

$$X_{P,n} = \{[x_i] : (x_i^n)_i \in P\}, \qquad G_{m,n} = (\mu_n)^m/\mu_n.$$

Characters are tuples $b \in (\mathbb{Z}/n)^m$ with $\sum b_i = 0$; their support is $S(b) = \{i : b_i \neq 0\}$. Let $I_i \subset G_{m,n}$ be the image of the ith coordinate μ_n, and $A_i = n^{-1}\sum_{g \in I_i} g^*$.

Theorem 3.1 (Exact support filter and census)**.** *For every $S \subset \{1, \dots, m\}$,*

$$E_S = \prod_{i \in S}(1 - A_i) \prod_{i \notin S} A_i \tag{3.1}$$

is a rational algebraic projector onto the sum of characters with support exactly S. These projectors are orthogonal and sum to the identity. For a fixed nonempty support of size s, the number of such characters is

$$a_s(n) = \frac{(n-1)^s + (n-1)(-1)^s}{n}. \tag{3.2}$$

The number of them of exact order $e \mid n$, $e > 1$, is

$$p_s(e) = \sum_{d \mid e} \mu(e/d) a_s(d), \qquad a_s(1) = 0. \tag{3.3}$$

They form $p_s(e)/\varphi(e)$ full rational orbits. Using the source's character-rank formula, the corresponding middle-degree ranks are $a_s(n)\binom{s-2}{d}$ and $p_s(e)\binom{s-2}{d}$ respectively, with zero when $s < d+2$.

Proof. On a character b, A_i is multiplication by $1_{b_i=0}$. Thus (3.1) has the asserted action, despite the relation among the coordinate subgroups. The commuting graph averages are genuine idempotents and products of complementary idempotents give orthogonality and the sum identity. To count nonzero entries with sum zero, average $\prod_i \sum_{b_i=1}^{n-1} \zeta_n^{tb_i}$ over t. For $t = 0$ the inner sum is $n-1$, and for every other t it is -1. This proves (3.2) even when n is composite. Tuples of order dividing e are exactly $(n/e)b'$ with $b' \in (\mathbb{Z}/e)^s$; support is preserved. Divisor inversion gives (3.3). A tuple of exact order e is fixed by a unit u only if $(u-1)b_i = 0$ for every i; its entries generate $\mathbb{Z}/e$, forcing $u = 1$. Orbits have size $\varphi(e)$. Finally apply the source rank formula character by character. □

The rank input is source Proposition `prop:characters` (2748–2807): it is obtained from the actual evaluation quotient and transport on a connected smooth family, not a count alone. Supports are not Galois orbits; a support can contain many such orbits and orders. The filter has denominator dividing n^m in its unreduced graph expression. It imposes exact membership before any period certificate.

Proposition 3.2 (Power descent)**.** *For $2 \le e \mid n$, the coordinate map*

$$q_{n,e} : X_{P,n} \longrightarrow X_{P,e}, \quad [x_i] \longmapsto [x_i^{n/e}]$$

is finite, surjective and is the quotient by a phase kernel of order $(n/e)^{m-1}$. Pullback identifies the b' character on $X_{P,e}$ with $b = (n/e)b'$ on $X_{P,n}$. Its inverse on that image is $(n/e)^{1-m}(q_{n,e})_*$. This includes zero-dimensional torsors.

Proof. Coordinate powers have no common zero; the defining equations are preserved. Taking coordinate roots proves surjectivity and finite phase fibers. On the dense locus with no zero coordinates the kernel action is free of the stated order. The smooth target is the finite quotient: either by normality and a finite birational comparison, or directly for a torsor. Quotient cohomology and the graph transfer give the assertions. □

4 Smaller sources for the actual Gale maps

We name the geometric input precisely. Write $s = |S| \ge d + 2$, $P_S = \mathrm{pr}_S P$, $Q_S = P_S^{\perp}$ for the ordinary *bilinear* dot product, and $h = s - d - 2$. Source Theorem `thm:gale` says that the algebraic map

$$f : X_{P_S,n} \times X_{Q_S,n} \longrightarrow F_n^{s-2}, \qquad (x, y) \longmapsto [x_i y_i],$$

pulls each full-support Fermat character line L_b to a perfect tensor in the two middle character spaces. We call this input **G**. The source proves it by its evaluation quotient, the exact wedge calculation, and transport of a morphism of local systems. We check the finite calculation below, but do not replace the actual geometric maps or their family transport by a rank test.

The graph $(x, y) \mapsto (f(x, y), y, x)$ has codimension $d + h$ in $(F_n^{d+h} \times X_{Q_S,n}) \times X_{P_S,n}$, source dimension $d + 2h$, and hence shift $-2h$. Its action on a witnessing tensor is

$$\alpha \otimes \beta \longmapsto (\mathrm{pr}_X)_*(f^*\alpha \smile \mathrm{pr}_Y^*\, \beta). \tag{4.1}$$

Here β must have the *opposite* character. In particular the middle source is the entire group $H^{d+2h}(F \times Y, \mathbb{Q})(h)$. Its witnessing subspace is $L_b \otimes H^h(Y, E)_{-b}(h)$. No algebraic Künneth projector is asserted or needed.

Theorem 4.1 (One point instead of a finite torsor)**.** *Suppose $s = d + 2$ and choose any point y of the finite dual root torsor $Y_S = X_{Q_S,n}$. All its coordinates are nonzero. Multiplication by y is an isomorphism*

$$f_y : X_{P_S,n} \xrightarrow{\sim} F_n^d.$$

*Consequently the rational algebraic correspondence $E_S p_S^* f_y^*$ from F_n^d to $X_{P,n}$ supplies every character of support S, simultaneously, from its entire degree $H^d(F_n^d, \mathbb{Q})$. It replaces the n^{s-1} point components of the source's finite dual torsor by one. It requires no input **G**.*

Proof. Q_S is a coordinate-general line, spanned by $(y_i^n)_i$. Thus P_S is the hyperplane $\sum y_i^n z_i = 0$. The diagonal automorphism $[x_i] \mapsto [y_i x_i]$ takes its root hypersurface exactly to $\sum z_i^n = 0$, with inverse multiplication by y_i^{-1}. The support quotient p_S is finite and its pullback identifies the desired characters, as in source 2792–2807. Both maps have algebraic graphs with coefficient 1; E_S enforces exact support rationally. An isomorphism supplies all of $H^d(X_{P_S,n})$, and hence the asserted subspace after pullback. The torsor has n^{s-1} points, but surjectivity onto its whole H^0 was never necessary to supply this target. This argument does not select one representative of a Galois orbit: it uses one actual complex point to supply all the characters at once. □

For codimension two there are at most m missing-one supports, so this replaces mn^{m-2} finite-source components by at most m Fermat sources; omit those with $a_{m-1}(n) = 0$. Different choices of y change the map by an algebraic phase automorphism, not its full image. A global choice of y over a parameter space can require a finite root base change. This is a fiberwise statement with rational cycle coefficients, not a descent assertion over a number field or the original coefficient torus. The source's use of all points is valid, but redundant for spanning.

Theorem 4.2 (Exact-order cyclic source for a rational orbit)**.** *Assume codimension two, $m = d+3$, and input $\boldsymbol{G}$ in degree e. Let $b \in (\mathbb{Z}/n)^m$ have full support and exact order e, so $b = (n/e)b'$ uniquely modulo n, with b' of exact order e. Put $C_e = X_{P^\perp,e}$ and $B_b = C_e/\ker \chi_{b'}$, a connected cyclic e-cover of $\mathbb{P}^1$. There is a rational algebraic map*

$$H^{d+2}(F_e^{d+1} \times B_b, \mathbb{Q})(1) \longrightarrow E_{\mathcal{O}(b)} H^d(X_{P,n}, \mathbb{Q}) \tag{4.2}$$

which is surjective, where $\mathcal{O}(b)$ is the full Galois orbit. The curve has genus

$$g(B_b) = 1 + \frac{e(m-2) - \sum_{i=1}^m \gcd(e, b_i')}{2}. \tag{4.3}$$

One curve works for the entire orbit. No assertion about Hodge classes or simple factors of its Jacobian is required for this map.

Proof. The curve quotient $r : C_e \to B_b$ is smooth and connected. Its residual character is faithful; its pullback identifies all $\chi_{ub'}$ eigenspaces for units u mod e, including their opposites. Compose $1 \otimes r^*$, the degree-e graph map (4.1), the power pullback $q_{n,e}^*$, and the rational orbit projector on the target. Every composition is an algebraic correspondence with total shift -2. After scalar extension, $\mathbf{G}$ and the perfect opposite-character pairing make it surjective onto every ub piece. The unit reduction $(\mathbb{Z}/n)^\times \to (\mathbb{Z}/e)^\times$ is surjective: choose a representative unit modulo each prime power of n, prescribing its reduction at the powers in e, and use the Chinese remainder theorem. Thus these are precisely the complete orbit of b. Rationality of the map and faithful scalar extension make its rational cokernel zero.

At branch label i, the residual local monodromy has order $e/\gcd(e, b_i')$. The ramification contribution is therefore $e - \gcd(e, b_i')$. Riemann–Hurwitz for a degree-e cover of the line gives (4.3). Primitivity of the whole tuple guarantees connectedness; individual entries need not be units. Finally primitive powers of a character have the same kernel, so they give the same curve. □

This is a supply reduction, not a claim that the scalar extension of a rational map has only one complex character image. With $n = e = 6, m = 4$ and $b' = (1, 1, 2, 2)$, the cyclic curve has genus 4, whereas C_6 has genus 145. The orbit's H^1 part has dimension $\varphi(6)(4-2) = 4$, not the full eight-dimensional $H^1(B_b)$. Formula (4.2) deliberately allows the entire source degree. Exact faithful projectors can remove the other rational blocks, but their existence does not certify Hodge genericity or algebraicity of arbitrary vectors.

4.1 The full-support Jacobian factor

Continue in codimension two and put $C = X_{P^\perp,n}$. The graph action of $G_{m,n}$ induces rational endomorphisms of its Jacobian; inverses of characters do not change support. The rational idempotent $E_{[m]}$ therefore determines an isogeny factor A_{full} of $\mathrm{Jac}(C)$. For precision, multiply the idempotent by an integer to obtain a genuine endomorphism, and take its connected image. The complementary idempotent gives a product decomposition up to isogeny; the addition map from the two images has an inverse in rational homomorphisms. Hence graph correspondences from A_{full} supply exactly

$E_{[m]}H^1(C)$. This is an actual algebraic abelian factor, not a chosen period subspace. Its dimension is

$$\dim A_{\text{full}} = \frac{(m-2)a_m(n)}{2}, \qquad g(C) = 1 + \frac{n^{m-2}(m(n-1)-2n)}{2}. \tag{4.4}$$

The first follows by summing the source rank $m-2$ over full-support characters and dividing by two; the second is Riemann–Hurwitz for the whole phase cover. A zero-dimensional factor means a point.

Lemma 4.3 (Bounded Fermat sources)**.** *Let $J_n = \mathrm{Jac}(F_n^1)$. The full cohomology of F_n^a, in every degree, is supplied by finitely many rational algebraic maps from powers J_n^v with $0 \le v \le a$ and the requisite Tate shifts. This uses algebraic domination and curve–Jacobian maps, not a Hodge conjecture theorem. The variety J_n is CM, as in the source.*

Proof. For $a = 0$ use points. For $a = 1$ the Abel map identifies degree one with $H^1(J_n)$; a point supplies the other degrees with shifts. For the inductive step write $a = 1 + b$ and choose $\varepsilon^n = -1$. The rational map

$$F_n^1 \times F_n^b \dashrightarrow F_n^{b+1}, \quad (x, y) \longmapsto (x_0 y_{b+1}, x_1 y_{b+1}, \varepsilon x_2 y_0, \ldots, \varepsilon x_2 y_b)$$

has the indicated Fermat image by summing powers. Where the first block has nonzero power sum, rescale that sum to -1 and set $x_2 = y_{b+1} = 1$ to construct preimages. It is dominant and generically finite. The base ideal is exactly (x_2, y_{b+1}), since at every base point some other x_i and y_j are units in local projective charts. Blowing up the smooth codimension-two center $F_n^0 \times F_n^{b-1}$ resolves it. The blowup cohomology is the product cohomology plus the center cohomology shifted by two, with algebraic maps: pullback from the product and exceptional-divisor pushforward from the center. The product uses at most $1 + b = a$ Jacobian factors by induction and the center at most $b - 1$. The resolved map f has generic degree $D > 0$, and $f_* f^* = D\,\mathrm{id}$, so pushforward supplies the target. All these operations compose actual cycles, and use full degrees via Künneth spanning, not algebraic Künneth projectors.

For CM, the phase character spaces of $H^1(F_n^1)$ have dimension at most one: the corresponding local system on the three-punctured line has rank $r - 2 \le 1$, and the invariant part is zero. The Hodge group commutes with the phase action and lies in its diagonal centralizer over a splitting field, a torus. This proves the asserted CM property. □

The domination and CM argument are retained from source 554–658, with the explicit exponent budget $v \le a$ extracted here. Historical Fermat domination is due to Shioda–Katsura; it is not a new theorem.

Theorem 4.4 (Reduced sufficient input for powers)**.** *For a fixed smooth diagonal intersection of two degree-n equations, of dimension $d \ge 1$, assume $\boldsymbol{G}$ and the source character-rank and support-quotient identifications. In every degree there is a finite rational algebraic supply from varieties*

$$A_{\text{full}}^{\epsilon} \times J_n^v, \qquad \epsilon \in \{0, 1\}, \quad 0 \le v \le d + 1, \tag{4.5}$$

with appropriate shifts. It follows that the single hypothesis

$$\mathrm{HC}((A_{\text{full}} \times J_n^{d+1})^u) \quad \textit{for every } u \ge 1 \tag{4.6}$$

is sufficient for $\mathrm{HC}(X_{P,n}^u)$ for every $u \ge 1$ at this fixed parameter. No assumption for arbitrary CM companions or the discarded isogeny factors of $\mathrm{Jac}(C)$ is needed in this implication.

Proof. Nontrivial characters have support m or $m-1$, since $m = d+3$. The latter use Theorem 4.1, followed by Lemma 4.3, with $v \le d$ and $\epsilon = 0$. For full support, the only witnessing curve subspaces in (4.1) are $H^1(C)_{-b}$ for full-support b. They are supplied from A_{full} by the isogeny-factor maps above. The Fermat factor has dimension $d+1$, so its supplies use $v \le d+1$. If its map has shift $2j$, composition with the Gale map gives shift $2(j-1)$ and is applied to the entire degree

$$H^{d+2-2j}(J_n^v \times A_{\mathrm{full}}, \mathbb{Q})(1-j) \longrightarrow H^d(X_{P,n}, \mathbb{Q}).$$

The required character Künneth pieces witness spanning. Applying the maps on whole rational degrees and then extending scalars leaves zero cokernel. Hyperplane cycles from a point supply invariant cohomology and every degree outside the middle.

For completeness we justify the Hodge transfer, rather than infer it from rank alone. Tensoring the finite supplies gives, by Künneth, surjections of polarizable rational Hodge structures onto each degree of X^u, with all Tate twists recorded. Any product of at most u varieties in (4.5) is an algebraic retract of $(A_{\mathrm{full}} \times J_n^{d+1})^u$: retain the required factors and insert origin points for a section. Thus (4.6) gives HC on every such product. A surjective morphism of polarizable rational pure Hodge structures admits a Hodge section: the kernel's orthogonal complement for a polarization gives a rational Hodge complement. A rational Hodge class on the target has a rational Hodge lift in the direct sum of twisted source degrees. Each component is algebraic by the hypothesis, and the actual correspondences push it to an algebraic target class. This proves the implication, without claiming that the abstract Hodge section is an algebraic correspondence. □

The hypothesis is genuinely smaller as a required list of varieties, but is not proved here. In particular HC on A_{full} and on J_n separately is insufficient: mixed powers can have additional Hodge classes. The source's full marked-variation theorem implies a stronger input at its specified generic points if its companion theorems are accepted. Our implication is valid at any fixed smooth parameter where (4.6) is independently known; it does not assert that all special parameters satisfy it.

4.2 Explicit sizes and the quadratic benchmark

For $(n, m) = (3, 5)$ one has $d = 2$, $g(C) = 55$, and $\dim A_{\mathrm{full}} = 15$. There are 10 full-support characters of rank 3 and 5 missing-one supports, each with 6 rank-one characters. The middle Betti number is $1 + 30 + 30 = 61$. The finite component count drops from $5 \cdot 3^3 = 135$ to 5, and the sufficient fixed product in (4.6) is $A_{\mathrm{full}} \times J_3^3$.

For $n = 2$, $a_s(2)$ is 1 for even s and 0 for odd s. If d is even then $m = d+3$ is odd, so $A_{\mathrm{full}} = 0$ and $\dim H^d(X) = m + 1 = d + 4$. All surviving nontrivial supports are the m missing-one supports. Since the Fermat quadric cohomology is algebraic (also from Lemma 4.3, since J_2 is a point), the one-point maps and hyperplane classes give algebraic cohomology in all degrees at every parameter. Products of this basis give HC on all powers. This is a rederivation of a known quadratic case, not a new special case credited to M064.

If d is odd, the only contributing tuple is $(1, \dots, 1)$ on even m. Theorem 4.2 uses the double cover branched at those m labels, of genus $(m-2)/2 = (d+1)/2$. For $m = 6$ this is genus 2, whereas the full phase curve has genus 17. The general sufficient condition reduces to HC on all powers of this hyperelliptic Jacobian, because J_2 is a point. This is compatible with the classical hyperelliptic correspondence, not its discovery; Laterveer's Theorem 2.1 gives a stronger Chow-motive formulation [4].

5 The signed Gale certificate and its limits

Proposition 5.1 (Exact wedge normalization)**.** *Let V have ordered basis $v_1, \dots, v_N$, and $a + b = N$. Use increasing wedge bases and the unnormalized alternating inclusion $\iota_j(v_1 \wedge \cdots \wedge v_j) = \sum_\sigma \operatorname{sgn}(\sigma) v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(j)}$. The unique map satisfying $(\iota_a \otimes \iota_b)\Delta = \iota_N$ is*

$$\Delta(v_1 \wedge \cdots \wedge v_N) = \sum_{|I|=a} (-1)^{\sum_{j=1}^{a}(i_j - j)} v_I \otimes v_{I^c}. \tag{5.1}$$

Its coefficient matrix is a signed permutation $J_{a,b}$, so $J_{a,b}^t J_{a,b} = I$ and $J_{b,a} = (-1)^{ab} J_{a,b}^t$. There is no factorial coefficient.

Proof. Every permutation of N labels has a unique first-block set I, followed by a permutation within I and one within I^c. Its sign is the shuffle sign times the two internal signs; the shuffle has $\sum(i_j - j)$ inversions. Expanding the right side under the two alternating inclusions therefore contains each of the $N!$ tensors exactly once, with its sign. Each set has one complement and the signs have square one, proving the isometry. Swapping the two blocks crosses ab pairs, proving the transpose formula, including $a = 0$ or $b = 0$. □

These are the source's conventions at its evaluation plane (2903–2926). If V has the basis-orthonormal positive metric, then $\|\iota_j(w)\|^2 = j!\,\|w\|^2$. Normalizing all three inclusions, including the top line, means imposing

$$\left(\frac{\iota_a}{\sqrt{a!}} \otimes \frac{\iota_b}{\sqrt{b!}}\right) \Delta_{\text{norm}} = \frac{\iota_N}{\sqrt{N!}}, \qquad \Delta_{\text{norm}} = \frac{J_{a,b}}{\sqrt{\binom{N}{a}}}.$$

Keeping the top inclusion unnormalized would instead give $\sqrt{a!b!}\,J_{a,b}$. These factors must not be confused with the unnormalized geometric identifications. Neither convention says that the cohomological polarization is a positive Euclidean error metric.

The evaluation-plane geometry uses distinct $t_1, \dots, t_m$ and $c_i = \prod_{j \neq i}(t_i - t_j)^{-1}$, for which $\sum_i c_i t_i^r = 0$ for $0 \le r \le m - 2$. Indeed interpolate z^r at the m points and compare the coefficient of z^{m-1}. These identities identify $P_a^\perp = \operatorname{diag}(c_i) P_b$; choosing nth roots of the c_i supplies actual coordinate scalings. The signed matrix calculation only applies after these geometric maps have been established. Our finite tests check both these rational moments and the wedge signs.

Corollary 5.2 (A normalization-aware local rank margin)**.** *In explicitly chosen frames suppose the true Gale matrix is $K = A J_{a,b} B$, where A, B are invertible frame matrices and the top-line generator is included in them. Then*

$$\sigma_{\min}(K) \ge \left(\left\|A^{-1}\right\|_2 \left\|B^{-1}\right\|_2\right)^{-1}.$$

Any implemented matrix $\widehat{K}$ with a rigorous operator error strictly below that bound is invertible. In the declared wedge frames $A = B = I$, error less than 1 suffices, and that threshold is sharp.

Proof. Apply $\|Jv\| = \|v\|$ and the lower bounds for A and B; then use the triangle inequality on $(K + (\widehat{K} - K))v$. At the endpoint remove one signed column of J, a perturbation of norm 1 giving a singular matrix. Strict inequality is therefore needed. □

This is a source-specific use of a standard singular-value inequality, not a new general perturbation theorem. Flat transport preserves the tensor and its rank, but may have arbitrarily bad conditioning in other frames; it does not provide uniform global unitary norms. The rational projectors above can force exact block membership of actual cycles. Small periods without that membership, a full-space test, and appropriate denominator information still cannot prove an algebraic relation.

6 Boundaries, comparison, and further work

The strongest local gain is removal of redundant source geometry, not relabeling a matrix. A missing-one dual torsor contributes one Fermat source, full support uses A_{full} instead of the entire Jacobian, and an exact-order orbit admits the lower-degree/cyclic supply (4.2). Equations (4.5)–(4.6) bound the auxiliary source powers and reduce a sufficient hypothesis. The standard character formula and the already-known quadratic case are benchmarks rather than independently new Hodge theorems.

The original source details retained here include purity of quotient cohomology with fixed points, the distinction between inverse and same characters in pairings, zero-rank determinants on a point, bilinear rather than Hermitian Gale orthogonality, nonunit branch entries, negative Gale shifts, whole rational source degrees, and the finite coordinate-root base change. No substantive correction to these source claims was established in this scoped work. The one-point and smaller factor constructions strengthen the bookkeeping without invalidating the larger source choices. Conversely, none proves source rank-one Lie classification, all-genus connected irreducibility, simultaneous scalar Hodge links, full marked-family comparison, or the companion CM Hodge theorem. The companions were not independently read for this report.

For general codimension $c = m - d - 1$, the same support formula gives $h_S = s - d - 2 \leq c - 1$. Conditional on $\mathbf{G}$, the original graph supplies cohomology from $F_n^{s-2} \times Y_S$ in degree $d + 2h_S$ with twist (h_S). Replacing the Fermat factors by the bounded supplies above proves a conditional all-powers implication if HC holds on all *mixed finite products* of the relevant Y_S and J_n powers. For $c = 3$ these dual sources have dimensions at most two. This is a concrete application of correspondence transfer, not an automatic HC extension: the dual root covers can themselves have high codimension, and the Gale involution is not a terminating descent algorithm.

Benefits and costs. Exact projectors prevent incomplete scalar-orbit certificates. The support/order census predicts dimensions before period computations. The one-point construction avoids exponentially many finite components, and cyclic quotients can sharply lower genus. On the other hand, graph coefficients can have large denominators, a rational block may remain large, and the cyclic refinement trades one common curve for several quotient curves. Point choices need not descend over a parameter space. Mixed-product HC, not separate HC, remains the relevant hypothesis.

Current primary literature, checked October 8, 2026. Schoen's 1988 paper supplies the historical phase-kernel/permutation construction, in a more restricted algebraicity setting [3]. The source's own broader determinant and genericity assertions are not validated merely by this precedent. Landesman–Litt–Sawin compute connected higher-Prym monodromy in a sufficiently high-genus range [5]; their abstract does not remove the source's low-genus work. Laterveer's introduction and Theorem 2.1 explicitly record the stronger known hyperelliptic Chow-motive relation for odd-dimensional intersections of two quadrics and the trivial even-dimensional case [4]. The September 28, 2026 Ji–Suzuki preprint studies Grothendieck-ring classes and Fano schemes for intersections of two quadrics [6]; its abstract is not a theorem about arbitrary-degree cover Jacobians. These literature checks were limited to the declared primary passages and metadata, not their complete proofs. Clay's current public problem page continues to list the general Hodge conjecture as unsolved [7]. No historical priority for the elementary refinements here is claimed.

Feasible next steps. Implement actual Chow-cycle representatives of the compressed maps in small cubic and quartic examples, keeping field-of-definition and cycle coefficient denominators

separate. Compute intersection matrices of their images, not merely Fourier ranks. For numerical work, enclose frame transport and period errors block by block using the original private report's complete perturbation budget. Test whether smaller isogeny factors admit independent mixed-power results at selected special parameters. Finally, review the geometric inputs $\mathbf{G}$ and the marked-variation and companion proofs separately; none is made redundant by a successful finite census.

7 Verification and contribution assessment

The accompanying standalone exact-arithmetic script checks the support and exact-order censuses against explicit tuples, the cyclic-kernel orbit grouping, Möbius subgroup projectors in rational group algebras, determinant block dimensions, signed wedge coefficients and transpose signs, rational Lagrange moments, and the displayed genus/rank examples. The author run passed 83,479 counted units: 34,631 explicit tuples, 46,233 alternating-expansion permutations, and 2,615 other checks in the separately listed categories. Its JSON output states all ranges. It uses no floating-point rank threshold and does not simulate geometric cycles, monodromy, or Hodge classes. Parent checks are separate evidence and are not included in author counts.

The assessment is C3, independently of catalogue applicability L3: Theorems 4.1, 4.2, and 4.4 give concrete source reductions, with full proofs of their local steps and explicit geometric premises. The other propositions provide exact membership, normalization and bounded verification. Neither textbook Fourier identities alone nor the number of finite tests earns this score. No new global Hodge theorem, general cycle-construction algorithm, or uniform period-conditioning result is claimed.

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